

Yousef Alsaqa

Mechatronics & Robotics Engineer building production software and hardware systems

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EDUCATION

B.A.Sc., Mechatronics & Robotics Engineering · Queen's University, Kingston, ON

Sep 2021 - Jun 2025

- Fourth-Year Dean's List (2025) · Principal's Scholarship for Academic Excellence (\$4,000)

WORK EXPERIENCE

Business Analyst · Queen's University, Undergraduate Admission & Recruitment

Jul 2026 - Present

- Finalized the document attacher into the office standard, root-causing a page-reload bug that was doubling its runtime. Used office-wide: **20,000–30,000 documents** per cycle.
- Grew the evaluation dashboard into a system driving **5 parallel** PeopleSoft sessions across multiple curricula; automated a CEGEP evaluator and an average-to-student matcher.
- Turned the Commerce-only admission analysis dashboard into a generic, config-driven system where a new program is a config file rather than new code; the Engineering analysis then plugged in as exactly that: ten years of data, high school average and the five required 4U courses against Year 1 GPA and each APSC course, split by school designation and admission basis.

Admission Coordinator · Queen's University, Undergraduate Admission & Recruitment

Jan 2026 - Jun 2026

- Built a Tkinter/Selenium tool that semi-automated the admission-decision workflow: read the documents, type one number, the tool does the rest. **~1,200 applications**.

Recruitment Representative · Queen's University, Undergraduate Admission & Recruitment

May 2025 - Jan 2026

- Built a fuzzy-matched location analyzer covering **5 years, ~170K applications**: feeder-school rankings, regional analysis, year-over-year conversion.
- Random Forest analysis over **10+ years** of Commerce admissions data, testing which admission inputs, high school average, subject grades, supplementary application, actually predict first-year GPA. The thresholds have since gone up.

Freelance Web Developer · Kingston, ON

Jul 2026 - Present

- Shipped production sites for two Kingston businesses: Next.js + Supabase, owner-editable menus and dashboards, deployed on Vercel.

Mechanical Engineering Intern · Dorsch Holding GmbH, Abu Dhabi, UAE

Jun 2023 - Aug 2023

- Redesigned a water drainage system from pumped to gravity-fed on the Bu Hasa oil-field expansion after fluid simulations exposed inefficiencies; reworked pipe layouts to cut pressure losses.

PROJECTS

World Cup Fantasy · full-stack platform, live through the entire World Cup 2026

Jan 2026 - Jul 2026

- Solo-built Next.js/TypeScript/PostgreSQL platform serving real users through the tournament; live-scoring engine reconciled against an error-prone external feed, verified with hand-rolled test harnesses and a live match first.
- Admin actions write to an immutable, revertible audit log; forked into a live La Liga product at ~80% reuse.

Vela · voice assistant with desktop control, in daily use

Aug 2026 - Present

- TypeScript agent on the Claude Agent SDK: wake-word voice, app/media/window control, watch-and-notify triggers. Memory is human-editable markdown in an Obsidian vault: one note per fact, hand corrections read back on recall.

CAST Medical Interface · RFID patient simulator, deployed at Queen's School of Nursing

Sep 2024 - May 2025

- Assistant lead, software & hardware integration (senior capstone): 12 RFID-tagged anatomical positions, 28 audio scenarios, Flask on a Raspberry Pi, **~\$250** of hardware replacing six-figure commercial simulators.

Queen's AutoDrive · Level 4 autonomous vehicle, SAE AutoDrive Challenge

Sep 2021 - May 2024

- CNN lane-perception optimized for embedded inference (TensorRT, Jetson Xavier NX); then system integration, synchronizing ROS2 fusion, planning and control over CAN at **10 Hz**.

Resonance · Unreal Engine 5.8 action game, C++ gameplay

Jul 2026 - Present

- Combat on driven-oscillator physics: probe a part, match its dampener frequency, then overload or stabilize it. Chaos destruction; every verb logs telemetry and feel is debugged from the numbers.

Formula SAE Steering System · Electrical Lead, custom driver telemetry

Sep 2022 - May 2023

- Designed a custom PCB and TFT display delivering real-time telemetry over CAN at **500 kbps**; root-caused termination-resistor issues behind intermittent bus dropout at speed.

TECHNICAL SKILLS

SOFTWARE & WEB

TypeScript, Next.js/React, Node.js, Python, C/C++, SQL/PostgreSQL, SQLite, Prisma, Supabase, Git

ROBOTICS & EMBEDDED

ROS/ROS2, OpenCV, SLAM, CAN bus, Arduino, Raspberry Pi, SolidWorks, MATLAB/Simulink, Unreal Engine 5

DATA & ML

Pandas, scikit-learn, PyTorch, Random Forest, TensorRT, rapidfuzz

AUTOMATION

Selenium, Playwright, Tkinter, YAML/JSON config, PeopleSoft, Vercel, Claude Agent SDK